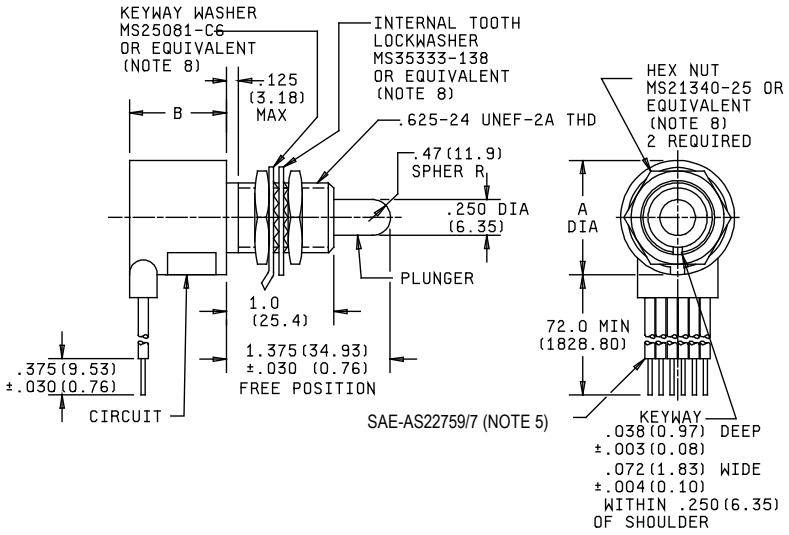
INCH-POUND
MIL-PRF-8805/80H
21 October 2019
SUPERSEDING
MIL-PRF-8805/80G
29 April 2009

PERFORMANCE SPECIFICATION SHEET

# SWITCH, SENSITIVE, LIMIT, HERMETIC SEAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-8805.



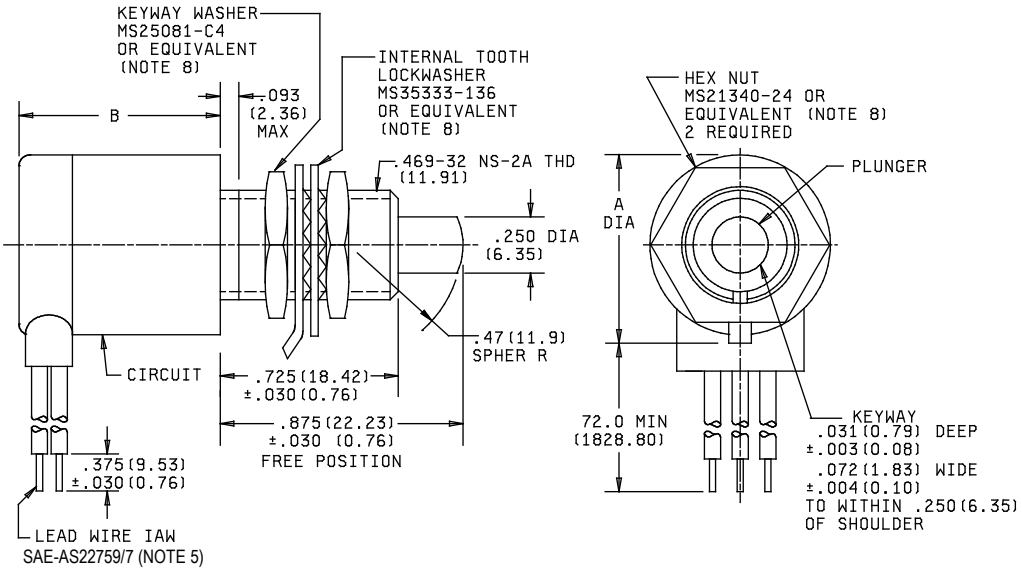

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.8 mm) for one place decimals and ±.02 (0.5 mm) for two place decimals, ±.005 (0.13 mm) for three place decimals
4. Configuration optional provided specified dimensions are not exceeded.
5. Lead wire shall be marked at intervals of 3.5 inches (89 mm) maximum with switch circuit identification number followed by wire gauge number (1-20). The marking shall be legible, permanent, and shall not impair the wire characteristics.
6. See table III for A and B dimensions.
7. MS21340-25 per MS21340. MS25081-C6 per MS25081. MS35333-138 per NASM35333.
8. Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for hexagon nut, lock washer and key washer material. Dimensions shall be in accordance with the referenced hardware specifications.

FIGURE 1. Pin plunger switch with .625 inch (15.88 mm) threaded bushing.



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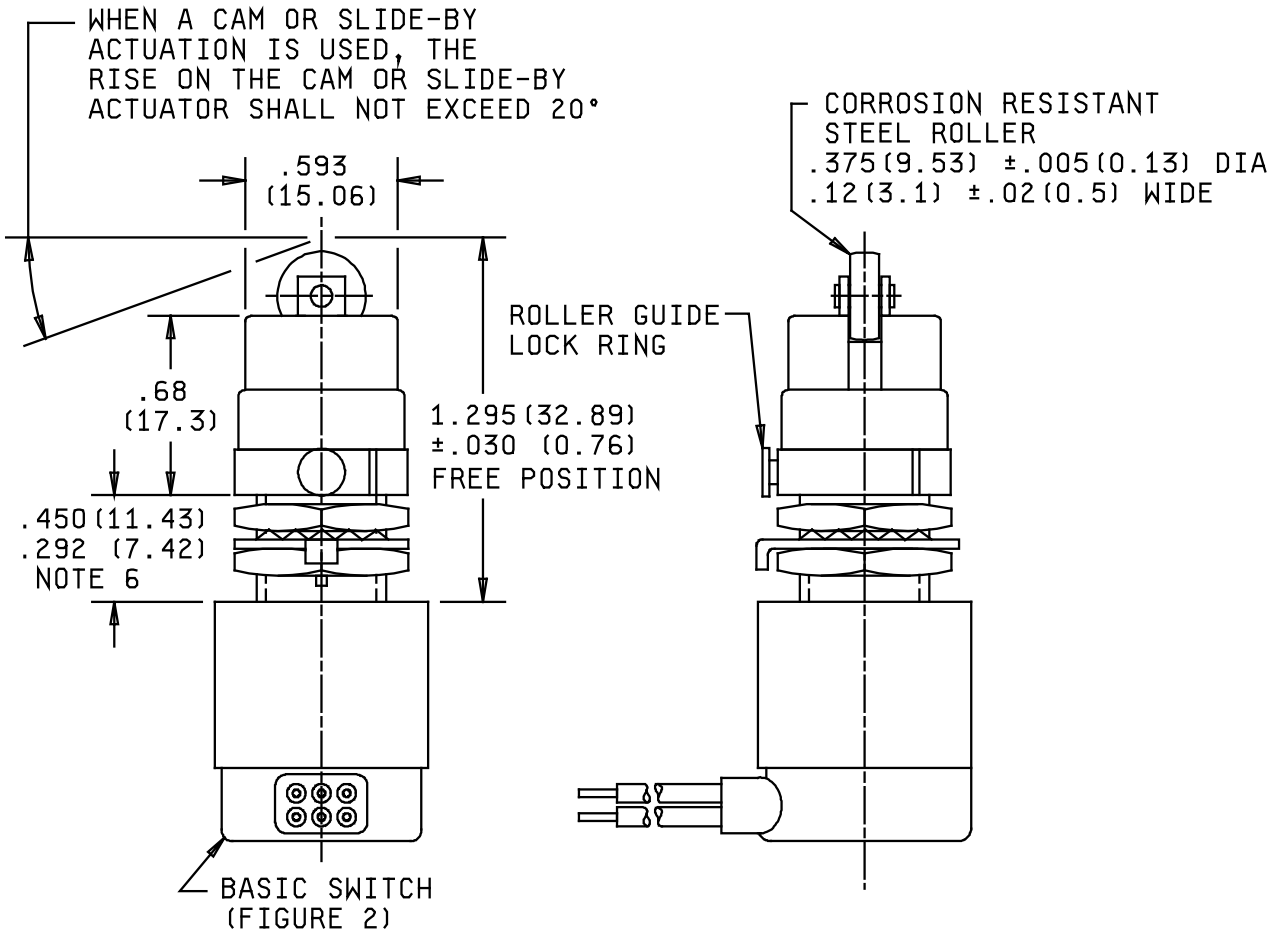

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.8 mm) for one place decimals and ±.02 (0.5 mm) for two place decimals, ±.005 (0.13 mm) for three place decimals
4. Configuration optional provided specified dimensions are not exceeded.
5. Lead wire shall be marked at intervals of 3.5 inches (89 mm) maximum with switch circuit identification number followed by wire gauge number (1-20). The marking shall be legible, permanent, and shall not impair the wire characteristics.
6. See table III for A and B dimensions.
7. MS21340-24 per MS21340. MS25081-C4 per MS25081. MS35333-136 per NASM35333.
8. Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for hexagon nut, lock washer and key washer material. Dimensions shall be in accordance with the referenced hardware specifications.

FIGURE 2. Pin plunger switch with .469 inch (11.91 mm) threaded bushing.



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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.8 mm) for one place decimals and ±.02 (0.5 mm) for two place decimals, ±.005 (0.13 mm) for three place decimals
4. Configuration optional provided specified dimensions are not exceeded.
5. Lead wire shall be marked at intervals of 3.5 inches (89 mm) maximum with switch circuit identification number followed by wire gauge number (1-20). The marking shall be legible, permanent, and shall not impair the wire characteristics.
6. Roller guide shall permit positioning and locking of plunger in increments of 45º azimuth. Roller guide location can vary between .292 (7.42 mm) and .450 (11.43 mm) while maintaining .125 (3.18 mm) minimum overtravel.

FIGURE 3. Roller plunger switch with .469 inch (11.91 mm) threaded bushing.



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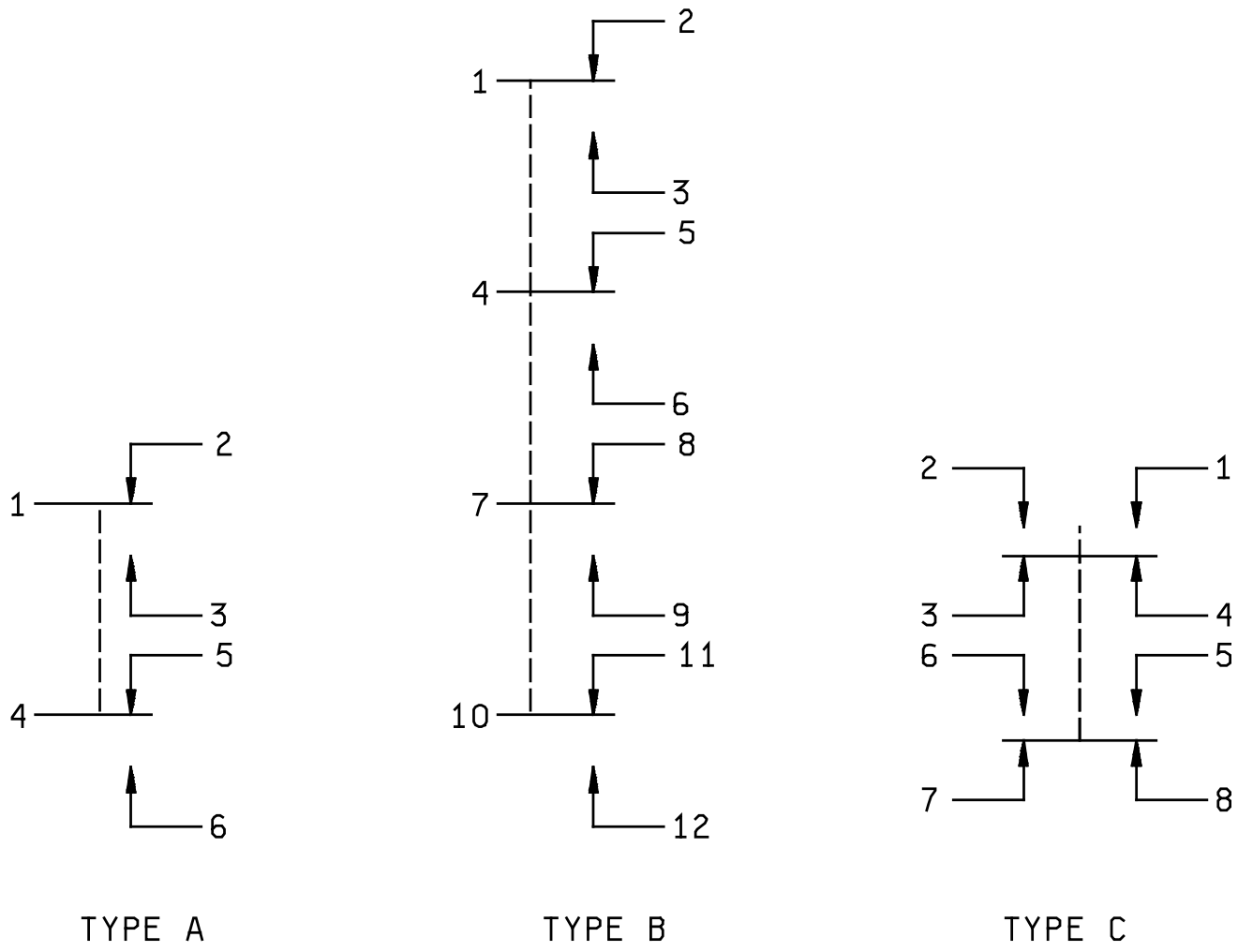



FIGURE 4. Circuit diagrams.



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REQUIREMENTS:

Dimensions and configuration: See figures 1 through 3.

Enclosure design: 5 (hermetic) switch chamber.

Temperature characteristic: 1 (-55°C to +125°C).

Shock type:

M8805/80-01, and M8805/80–03, through M8805/80–08: MIL-STD-202- 213, half-sine shock pulse of 200 g amplitude with .007 second duration.
M8805/80-02:M (100 g)

Vibration grade:

M8805/80-01, and M8805/80–03 through M8805/80–08: 2 (10-2000 Hz except at 20 G's).
M8805/80-02: 1 (10 to 500 Hz).

Weight with leads:

M8805/80-01, M8805/80-03, and M8805/80–07: 7.5 ounces maximum.
M8805/80-02 and M8805/80–08: 13.5 ounces maximum.
M8805/80-04 and M8805/80–06: 12.5 ounces maximum.
M8805/80-05: 6.5 ounces maximum.

Operating characteristics: 1/

Operating force:

M8805/80-01and M8805/80-02: 9 ±3 pounds.
M8805/80-03 through M8805/80–08: 8 pounds 8 ounces ±3 pounds 8 ounces.

Full overtravel force: 30 pounds maximum.

Release force:

M8805/80-01 and M8805/80–02: 4 pounds minimum.
M8805/80-03 through M8805/80–08: 3 pounds minimum.

Pretravel:

M8805/80-01 and M8805/80–03 through M8805/80–08: .040 (1.02 mm) maximum.
M8805/80-02: .070 (1.78 mm) maximum.

1/ ±20 percent variation from initial values acceptable after test.



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Movement differential:

M8805/80-01 and M8805/80-03 through M8805/80–08: .020 (0.51 mm) maximum.
M8805/80-02: .045 (1.14 mm) maximum.

Overtravel:

M8805/80-01 through M8805/80–04: .250 (6.35 mm) minimum.
M8805/80-05 through M8805/80–08: .125 (3.18 mm) minimum.

Coincidence of operating and releasing points: .010 (0.25 mm) maximum.

Finish: Switch housing shall be processed to resist corrosion.

Plastic: M8805/80-02 may use general purpose mineral and flock filled phenolic internally.

Strength of actuating means: 250 pounds.

Strength of mounting bushing:

M8805/80-01 through M8805/80-04: 400 pound-inches.
M8805/80-05 through M8805/80-08: 200 pound-inches.

Contact resistance: Not applicable.

Dielectric withstanding voltage:

Sea level: 1,000 V rms. In qualification inspection table after electrical endurance, the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable.

Fluid resistance: Except for the cut of the lead wire, switches shall be submerged in each of the following fluids for 2 to 2.5 minutes, which shall consist of one cycle (one cycle is 10 to 12.5 minutes total). Each switch shall be subjected to three cycles.

a. Turbine fuel (MIL-DTL-5624).

b. Hydraulic fluid (SAE AS1241).

c. Coolanol 1/ (MIL-PRF-87252).

d. Ethylene glycol (ASTM-E1119).

e. Lubricating oil (MIL-PRF-7808)

After each immersion, the excess fluid shall be blown off the external surfaces of the switch with an air jet. Following the third cycle, the switch shall be subjected to and shall meet the requirements for dielectric withstanding voltage, insulation resistance, operating characteristics, seal, and marking.

1/ Monsanto Company registered trademark.



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Seal test: Only watertight test shall be performed in group A.

Sand and dust test: Applicable.

Icing test:

M8805/80-01: Immersion cycles –3.
M8805/80-02 through M8805/80–08: Not applicable

Intermediate current:

M8805/80-01: 100,000 cycles.
M8805/80-02 through M8805/80–08: Not applicable.

Mechanical endurance: 100,000 cycles.

Electrical endurance: 25,000 cycles.

Electrical ratings: See table I.

Qualification group submission: See table II.

Part or Identifying Number (PIN): See table III.

TABLE I. Electrical ratings.

| Load | 28 V dc sea level and 100,000 ft. | | |
|---|---|---|---|
| | -01 | -02 | -03 through –08 |
| Resistive<br>Inductive<br>Motor | (Amperes)<br>5<br>3<br>4 | (Amperes)<br>10<br>3<br>6 | (Amperes)<br>7<br>4<br>4 |

TABLE II. Qualification inspection group submission.

| Examination or test | Basic switch | Other switch samples | Extent of approval |
|---|---|---|---|
| Qualification inspection table of MIL-PRF-8805 | M8805/80-01 | M8805/80-02<br>M8805/80-04<br>2 samples each for each electrical test.<br>M8805/80-02<br>M8805/80-03<br>M8805/80-04<br>M8805/80-06<br>M8805/80-07<br>1 each shall be submitted to the following test sequence.<br>1. Strength of actuator means<br>2. Vibration<br>3. Shock<br>4. Operating characteristics | All |



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TABLE III. Part or Identifying Numbers (PIN's).

| PIN M8805/80- | Figure | "A" ±.030 (.76) | "B" max | Number of lead wires | Lead wire size | Circuitry type (see figure 4) |
|---|---|---|---|---|---|---|
| 01 | 1 | 1.000 (25.4) | .980 (24.89) | 6 | 20 AWG | A |
| 02 | 1 | 1.500 (38.1) | 1.700 (43.18) | 8 | 18 AWG | C |
| 03 | 1 | .688 (17.48) | .980 (24.89) | 6 | 20 AWG | A |
| 04 | 1 | 1.000 (25.4) | 1.500 (30.48) | 12 | 20 AWG | B |
| 05 | 2 | .688 (17.48) | 1.100 (26.67) | 6 | 20 AWG | A |
| 06 | 2 | 1.000 (25.4) | 1.250 (31.75) | 12 | 20 AWG | B |
| 07 | 3 | .688 (17.48) | 1.100 (26.67) | 6 | 20 AWG | A |
| 08 | 3 | 1.000 (25.4) | 1.250 (31.75) | 12 | 20 AWG | B |

Referenced documents. In addition to MIL-PRF-8805, this document references the following:

| | |
|---|---|
| MIL-DTL-5624 | ASTM-E1119 |
| MIL-PRF-7808 | SAE-AS1241 |
| MIL-PRF-87252 | SAE-AS22759/7 |
| MIL-STD-202-213 | MS21340 |
| MS25081 | NASM35333 |

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Navy – EC
Air Force – 85
DLA –CC

Preparing activity:
DLA – CC

(Project 5930-2019-058)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .



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